Graph Theory

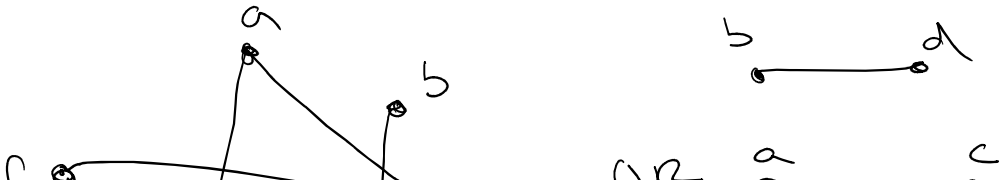
What is a graph?

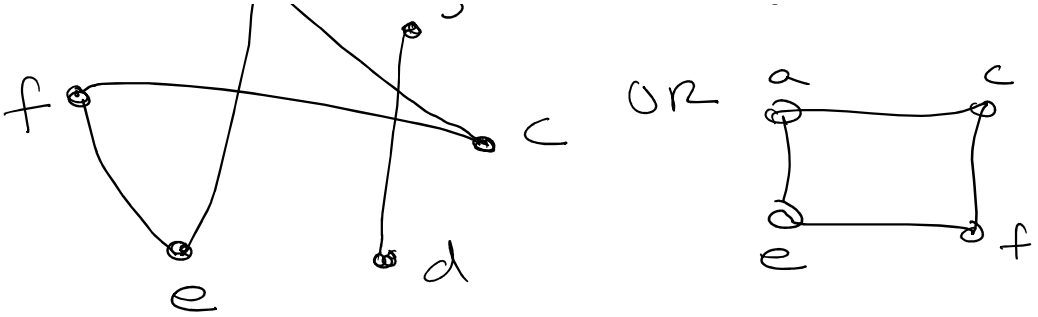
A graph G is a tuple, i.e., $G := (V, E)$, where V (or $V(G)$) is a set of elements called the vertex set (or node set) (each element in this set is called a vertex (or node)) and E (or $E(G)$) is a collection of 2-sized subsets of V and is called the edge set (each element in E is called an edge).

Example

$G = (V, E)$, where

$$\begin{cases} V = \{a, b, c, d, e, f\} \\ E = \{\{a, c\}, \{a, e\}, \{b, d\}, \{e, f\}, \{c, f\}\} \end{cases}$$





A more general notion

Hypergraph → is a tuple

$H = (V, E)$

V → vertex set (a collection of elements)

E → a collection of subsets of V.

(a subset in E is called a hyperedge)

Example $H = (V, E)$

$V = \{a, b, c, d, e, f\}$

$E = \{\{a, b, c, d\}, \{c, e, f\}, \{a, e, f\}, \{b, d\}\}$

Some definitions on graphs

$G = (V, E)$

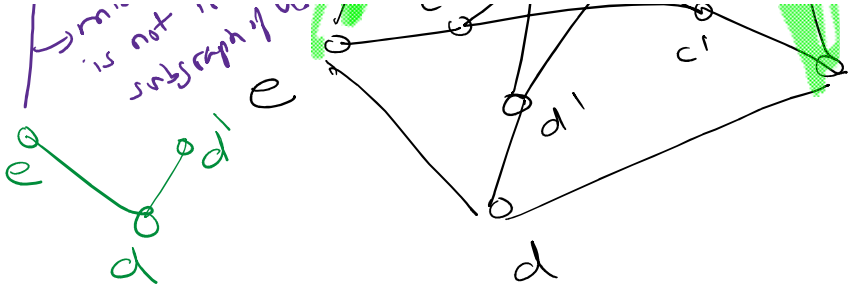
$V = \{a, b, c, d, e, a', b', c', d', e'\}$

$E = \{\{a, b\}, \{b, c\}, \{c, d\}, \ldots$

$H = (V', E')$

a b

missing. So H is not induced subgraph of G



$E = \{\{a,b\}, \{b,c\}, \{c,d\}, \{d,e\}, \{e,c\},$
OR $\{a,a'\}, \{b,b'\}$
OR $\{e,e'\}, \{a',c'\}, \{c',e'\}, \{e',b'\}, \{b',d'\}, \{d',a'\}\}$

$|V| = 10$

$|E| = 15$

a is a neighbour of b

$N_G(a) = \{b, a', e\}$

Notions

(i) degree of a vertex $\longrightarrow$ no. of edges it is present in.

$$\deg_G(a) = 3, \quad \deg_G(b) = 3$$

(ii) maximum degree of G, denoted by $\Delta(G)$, is

$$\Delta(G) := \max\{\deg_G(v) : v \in V(G)\}$$

above example, $\Delta(G) = 3$.

(iii) minimum degree of G, denoted by

$$\delta(G) := \min\{\deg_G(v) : v \in V(G)\}$$

above example, $\delta(G) = 3$

(iv) Given two vertices $u, v \in V$

(iv) given two vertices u and v, we say u is adjacent to v if $\{u,v\} \in E(G)$.

u is a neighbor of v in G if $\{u,v\} \in E(G)$.

(v) Neighborhood of a vertex v in G is denoted $N_G(v)$ and is defined as $N_G(v) = \{u : \{u,v\} \in E(G)\}$

(vi) $H = (V', E')$ is a subgraph of $G = (V, E)$ if $V' \subseteq V$ and $E' \subseteq E$.

(vii) a vertex v is incident with an edge e if $v \in e$.
↳ 2-sized subset of V.

(ix) $H = (V', E')$ is an induced subgraph of $G = (V, E)$ if $V' \subseteq V$ and

of $G=(V,E)$ if $V' \subseteq V$ and

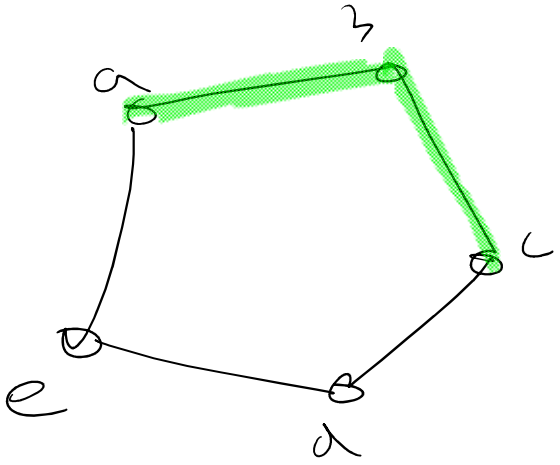
$E' = \{(u,v) \in E(G) : u,v \in V'\}$

(x) $H=(V',E')$ is a spanning subgraph of $G=(V,E)$ if H is a subgraph of G and $V'=V$.

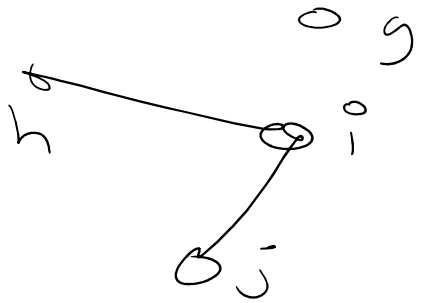
↳ spanning

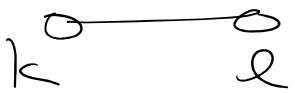
(xi) Average degree of a graph G, denoted $d_{avg}(G)$ is

$$d_{avg}(G) := \frac{\sum_{v \in V(G)} deg_G(v)}{|V|}$$



for $H=(V,E)$



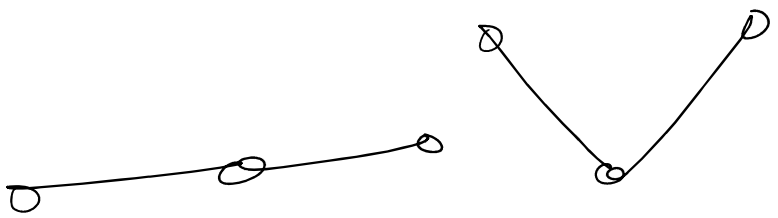


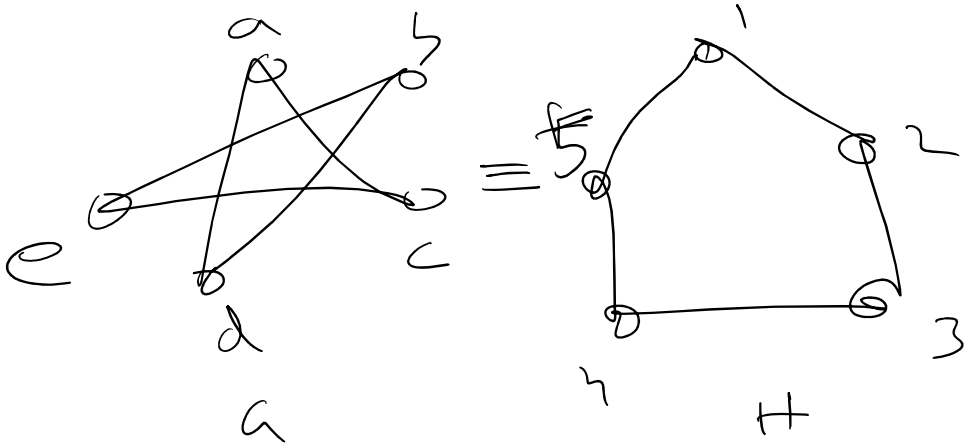
$V(H) = \{a, b, \ldots, l\}$

$E(H) = \{\{a,e\}, \{a,b\} \ldots \{k,l\}\}$

* sum of degrees $= 2 \cdot |E(G)|$.

Therefore no. of odd degree vertices is even.





G is isomorphic to H.

$f : V(G) \to V(G)$ as

$f(a) = 1, f(c) = 2, f(e) = 3, f(b) = 4,$

$f(d) = 5$

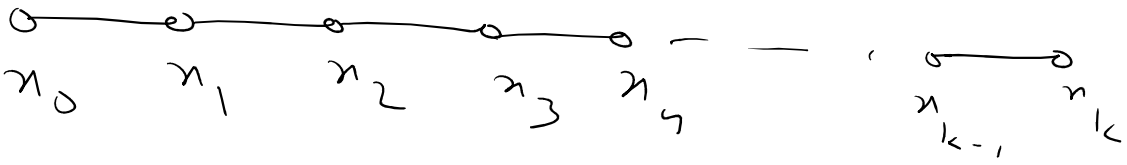
* A path is a non-empty graph

$P = (V, E)$, where

$V = \{x_0, x_1, x_2, \ldots, x_k\}$

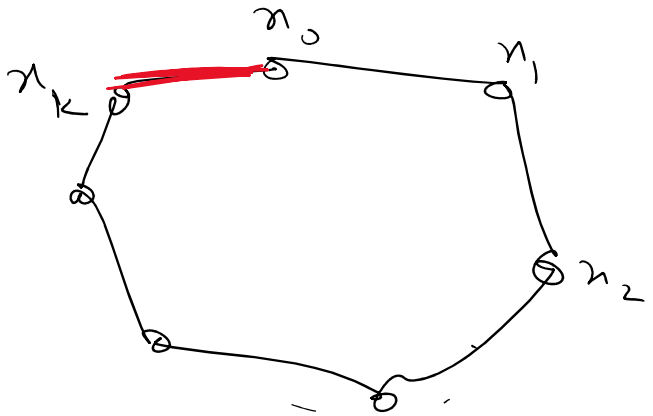
$E = \{\{x_0, x_1\}, \{x_1, x_2\}, \{x_2, x_3\}, \ldots$

$\ldots \{x_{k-1}, x_k\}\}$



Length of a path is the no. of edges in it.

* Cycle



Let $P = (V, E)$ be a path, where $V = \{x_0, x_1, \ldots, x_k\}$

$E = \{\{x_0, x_1\}, \{x_1, x_2\}, \ldots,$

$E = \{\{x_0, x_1\}, \{x_1, x_2\}, \ldots, \{x_{k-1}, x_k\}\}$

Then, $P + \{x_k, x_0\}$ is a cycle

Lemma Every graph has
(i) a path of length $\delta(G)$, and (starting at a vertex with min degree)
(ii) a cycle of length $\delta(G) + 1$. $(\delta(G) \geq 2)$

Proof.

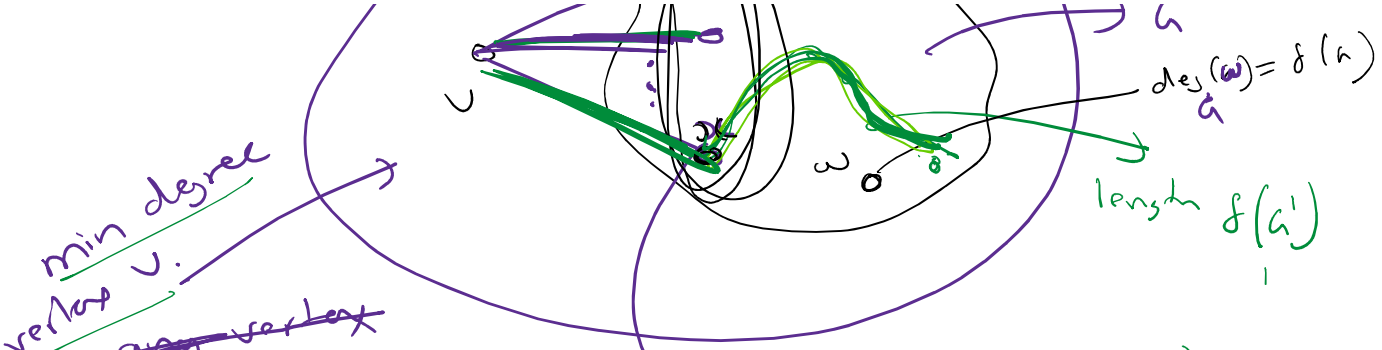
Proving (i) Consider a graph G with min degree $\delta(G)$.
By mathematical induction on no. of vertices.

Base case: $|V(G)| = |V| = 1$.

$\delta(G) = 0$.
0-length path

Induction Step. Assume the stmt is TRUE for all graphs on n (at most) vertices.
Suppose G has $n+1$ vertices.





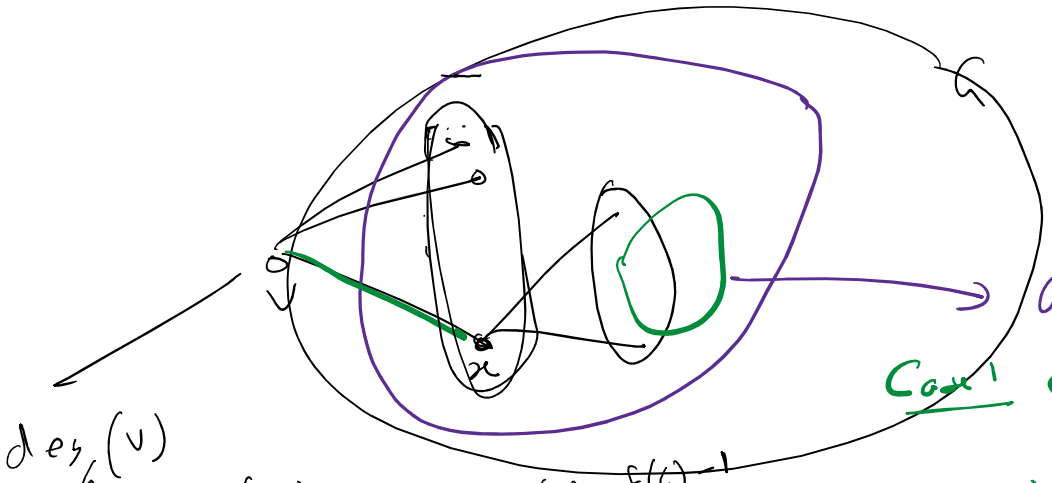
Case 1: $\delta(G') \geq \delta(G)$

good.

Case 2 $\delta(G') < \delta(G)$

Then,

$\delta(G') = \delta(G) - 1$



Case 1 $\delta(G') \geq \delta(G)$.

good

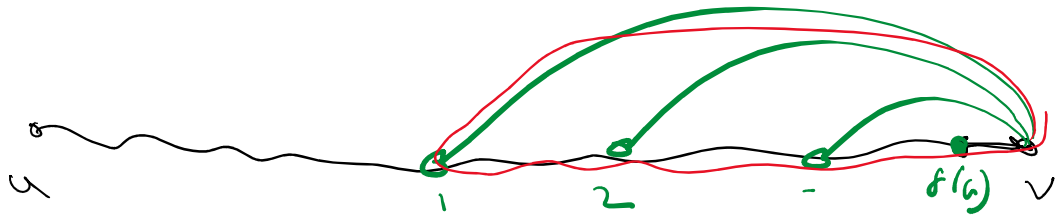
Case 2 $\delta(G') = \delta(G) - 1$

↳ How to go forward??

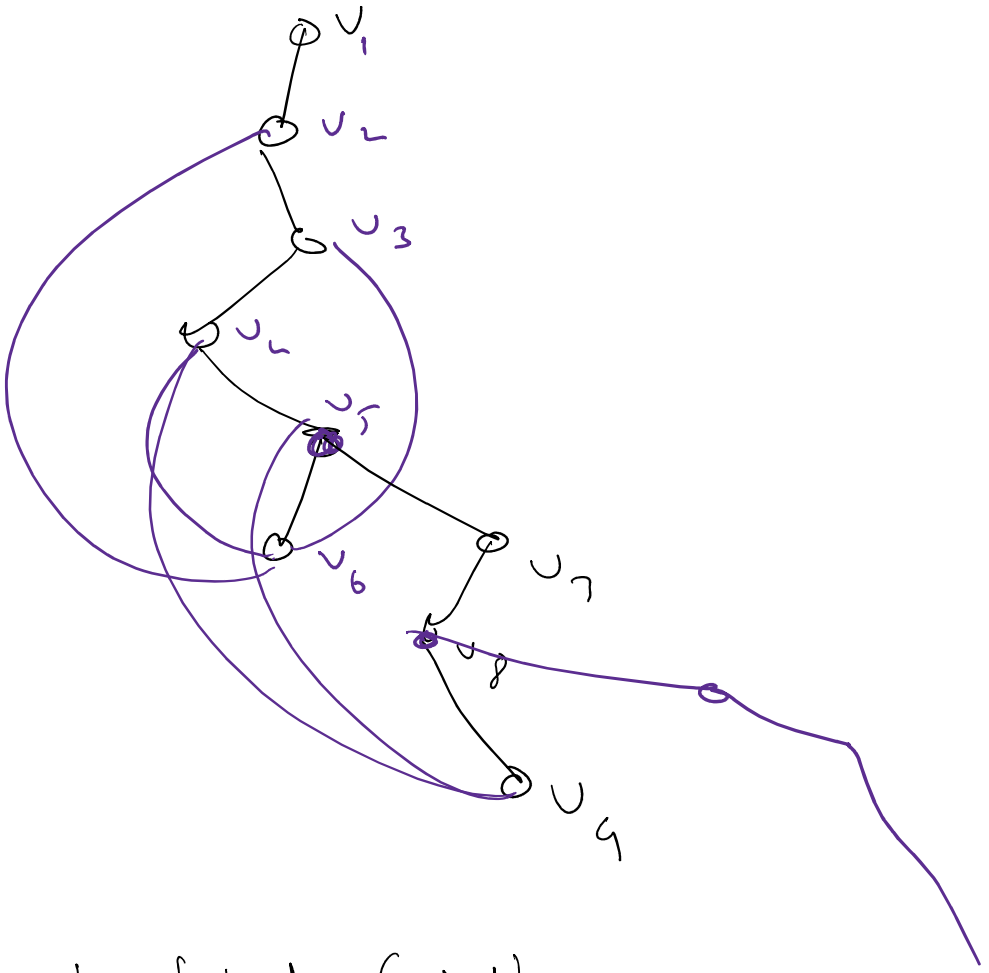
Alternate approach

Proving (i) and (ii)

Take the longest path in G.



$\deg_G(v) \geq \delta(G)$



Lecture by Subrata (Wed)

Thursday

Recap:

radius, diameter, central vertex, girth, circumferance, distance.

Stmt. Given a graph G, of G.

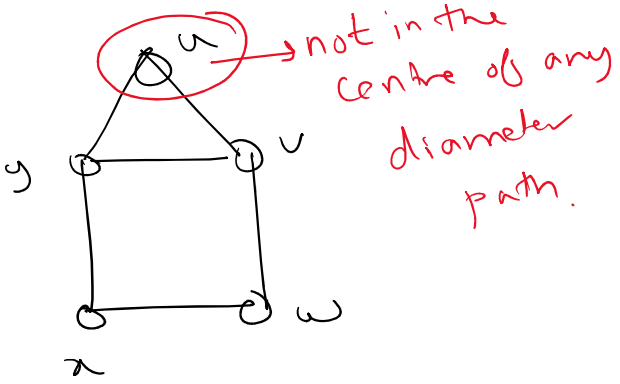
(1) Every central vertex is in the centre

graph u, of G.

(i) Every central vertex is in the centre of a diameter path

(ii) The centre of a diameter path is a central vertex of G.

Statement (i) is wrong

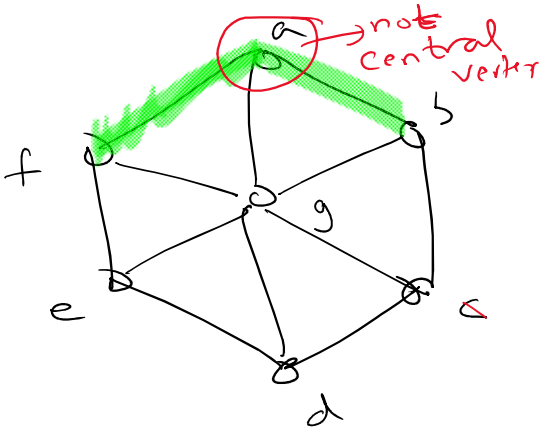


radius $= 2$

diameter $= 2$

every vertex is a central vertex.

Statement (ii) is wrong



radius $= 1$

diameter $= 2$

g is a central vertex of G.

Proposition.

$$\text{rad}(G) \leq \text{diam}(G) \leq 2 \cdot \text{rad}(G).$$

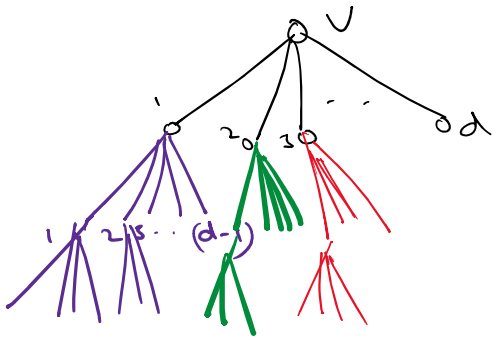
Proposition $\quad \text{radius}(G) \leq k$

max deg of $G \leq d$, where $d \geq 3$

Show that the graph G has fewer than

$$\frac{d}{(d-2)}(d-1)^k \text{ vertices.}$$

Proof: Let v be a central vertex of G.



Level 0 → 1

Level 1 → d

Level 2 → $d(d-1)$

Level 3 → $d(d-1)^2$

Level k → $d(d-1)^{k-1}$

$$\#\text{ vertices} \leq 1 + d + d(d-1) + \cdots + d(d-1)^{k-1}$$

$$\leq 1 + d\left(1 + (d-1) + \cdots + (d-1)^{k-1}\right)$$

$$= 1 + d\left(\frac{(d-1)^k - 1}{d-2}\right)$$

$$= \frac{d(d-1)^k}{d-2} - \frac{d}{d-2} + 1$$

$$< \frac{d(d-1)^k}{d-2}$$

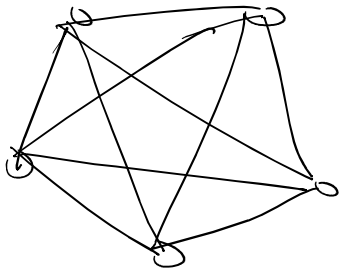
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Some commonly seen graphs

(i) Complete graph on n vertices, denoted K_n, has all the $\binom{n}{2}$ edges

K_n, and it has all the $\binom{n}{2}$ edges.

K_5



Some definitions

* Independent set: Given a graph G, $S \subseteq V(G)$ is called an independent set if no two vertices in S have an edge between them.

Independence no. of a graph G, denoted $\alpha(G)$ is the size/cardinality of a largest independent set in G.

* Clique: A set $S \subseteq V(G)$ is a clique if every two vertices in S have an edge between them in G.

Size/cardinality of a largest clique in G is called clique number of G. It is denoted $\omega(G)$.

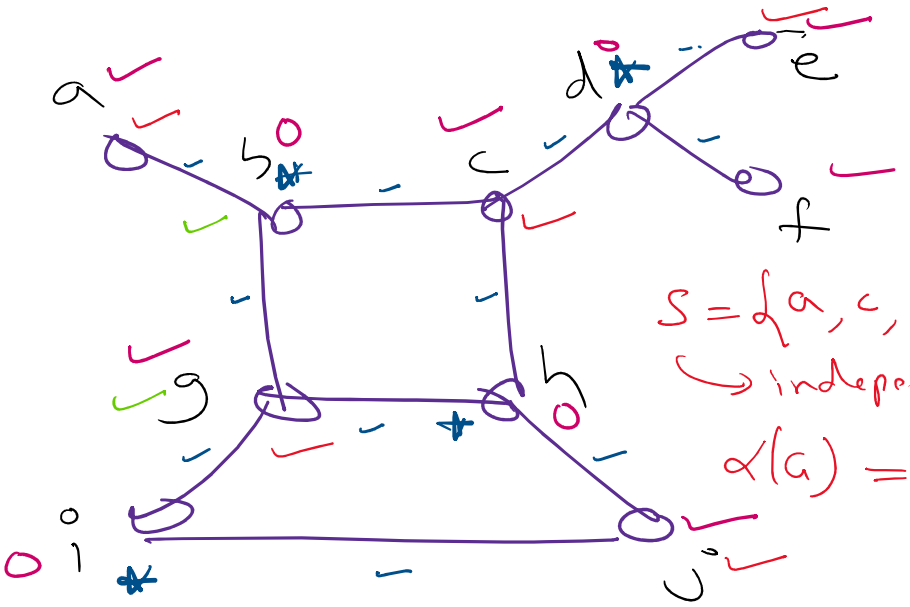
* Vertex cover : A set $S \subseteq V(G)$ is a vertex cover of G if every edge in G has an endpoint in S.

A smallest cardinality vertex cover of G is called a minimum vertex cover

* Dominating set : A set $S \subseteq V(G)$ is called a dominating set of G if every vertex in G is either in S or has a neighbor in S.

Example to explain the notions





$S = \{a, c, e, g, j\}$ → independent set

$\alpha(G) = 6$

$S' = \{b, g\}$ is a clique.

$S'' = \{b, d, h, i\}$ is a vertex cover.

Is it the smallest one?

$S''' = \{b, d, h, i\}$ is a dominating set.

Is this a smallest dominating set?

→ Coming back to examples of common graph

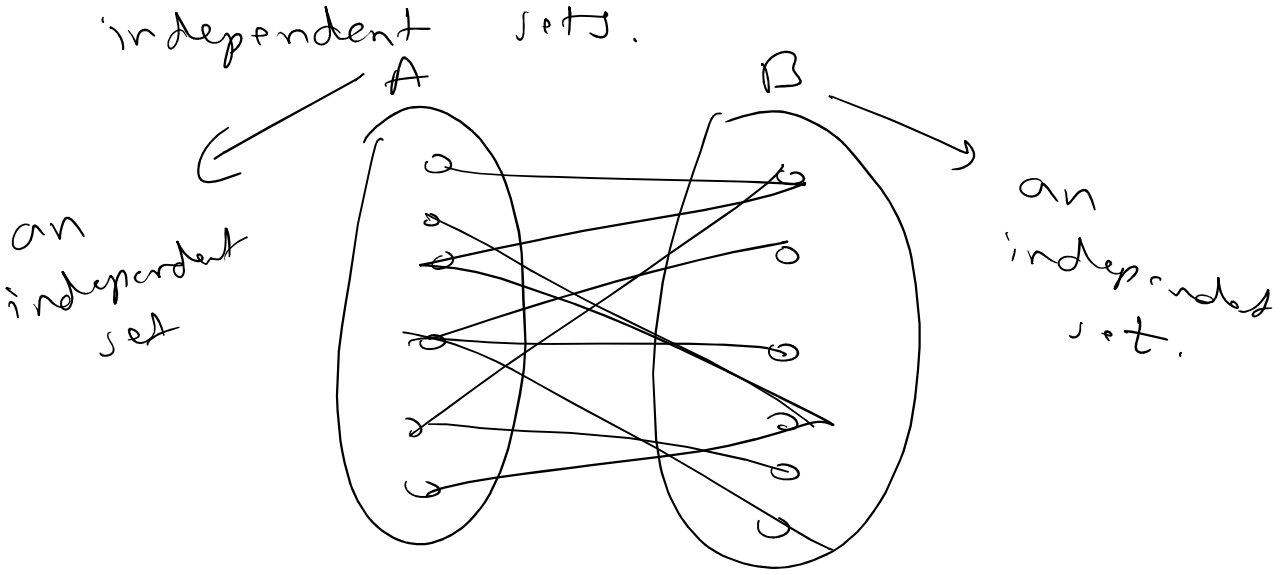
$\alpha(G)$	→ max indep. set size	
$\beta(G)$	→ min vertex cover size	
$\gamma(G)$	→ largest dominating set size.	

(i) Complete graph on n vertices

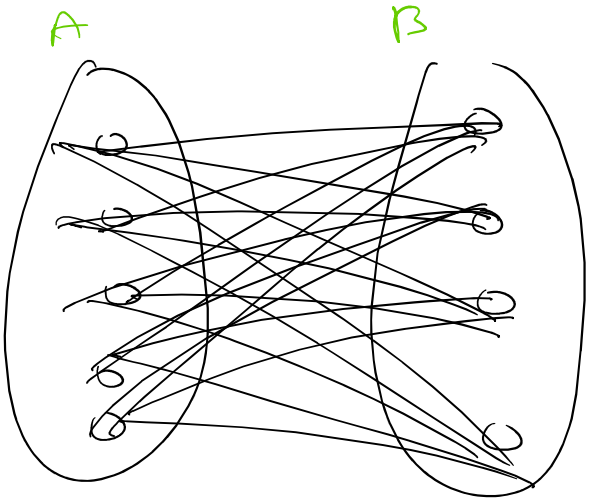
$\alpha(G) = 1$, $\beta(G) = n-1$, $\gamma(G) = 1$

(ii) Bipartite graph:

A graph whose vertices can be partitioned into two



(iii) Complete bipartite graph



$\alpha(G) = \max(|A|, |B|)$

$\beta(G) = \min(|A|, |B|)$

$\gamma(G) = 2$

(iv) Path graph on n vertices



(v) Cycle graph on n vertices

